

Can News Content Predict the Next Day’s Market Movement?

A Test Based on U.S. AI Technology Stocks

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Abstract

Since 2023, AI-related topics have gained increasing momentum in both investment and academic circles, with fluctuations in AI technology stocks becoming a hot topic. News about these companies also carries a certain amount of information. This paper investigates whether, under constraints of limited authoritative reporting and low signal-to-noise ratio, modeling news can predict the next-day price direction of individual stocks and consistently outperform baseline benchmarks. We selected five U.S. tech stocks (covering Bloomberg news from 2021 to 2025, approximately 13,106 articles), three strongly associated with AI and two weakly associated. First, we transformed news into features using a two-step NLP pipeline: FinBERT for sentiment scoring and BART for zero-shot narrative classification. Then, we introduced NG-LP (Narrative-Gated Local Projections) — a method that combines local projections with dual gating mechanisms and daily majority voting. Results show that in out-of-sample predictions for 2025, NG-LP significantly outperformed the baseline, while complex machine learning models (decision trees, SVM, LSTM) and logistic regression failed to pass statistical tests due to overfitting on small samples. Under these conditions, simple linear models enhanced with narrative gating outperformed more complex models. The combined out-of-sample accuracy for the three core AI stocks (NVDA/MSFT/META) reached 64.5%, surpassing the 53.5% baseline (+11.0 percentage points, $P(\text{acc} > \text{base}) = 99\%$). In contrast, GOOGL, a stock with weaker AI ties, did not outperform the baseline, indicating that this signal is specific to AI narratives rather than a general market effect. Prediction accuracy exhibited a U-shaped distribution relative to volatility, performing best on calm days and during shock events. These findings suggest that news narratives carry lagged directional information not yet fully priced in, challenging the semi-strong form of the efficient market hypothesis. However, our conclusions are limited to directional predictability and do not imply tradable excess returns.

1 Introduction

Since 2023, AI has rapidly become a hot topic in the investment market, with related news emerging constantly. However, predicting stock price movements based solely on news is not easy. The challenge lies in the structure of the information itself: the more authoritative a report, the more neutral and restrained its language tends to be, resulting in weaker emotional signals that can be extracted. Compounded by the inherently limited availability of data samples, prediction faces a double constraint of weak signals and scarce data. Can we then design a

model capable of determining the direction of stock price movements from news content — even under these constraints — and consistently outperforming a baseline? If so, it would imply that there exists exploitable delayed transmission within news, an implication at odds with the semi-strong form of market efficiency — according to which market prices fully reflect all publicly available information, news is immediately priced upon release, and no exploitable lag exists. This is precisely the central question this paper seeks to address.

The foundation of this issue lies in Fama’s (1970) Efficient Market Hypothesis. In its semi-strong form, asset prices fully reflect all publicly available information, meaning any news is instantly and completely priced upon release, leaving no systematic exploitable lag. Around this proposition, a body of empirical literature has long questioned whether news texts still contain predictable information: Tetlock (2007) was among the first to show that media sentiment can predict stock returns, although the explained variance is typically low. Recently, large language models have advanced this inquiry significantly — Lopez-Lira & Tang (2026) found that evaluation scores assigned by ChatGPT/GPT-4 to news headlines positively predict return drift following news releases, particularly for small-cap stocks and negative news, though the fastest initial market reactions are not tradable; Chen, Kelly & Xiu (2023) further demonstrated that text representations extracted by large language models can improve predictions of expected returns; recent applied studies have also directly compared the performance of methods such as FinBERT, GPT-4, and logistic regression in stock price forecasting (Shobayo et al., 2024). However, most of these studies remain at the aggregate sentiment- and portfolio-level: they compress news into a single positive or negative sentiment, then construct cross-sectional portfolios across numerous stocks, relying on diversification effects across equities. They rarely address a more constrained and practically relevant question: when sentiment signals from authoritative reports are inherently weak, does news still carry exploitable directional information at the individual stock level with limited sample sizes? This paper fills precisely this gap.

To address the above issues, we first need to convert unstructured news text into numerical representations suitable for statistical models — a step whose quality determines the upper limit of all subsequent analyses. This paper employs a two-step pre-trained language model pipeline, assigning each model to do only what it does best: sentiment polarity is determined by FinBERT (Araci, 2019), a BERT model fine-tuned on financial corpora capable of accurately interpreting financial terms such as “beat,” “miss,” and “downgrade” along with their negations; narrative classification is handled by BART’s zero-shot classification (Lewis et al., 2020; Yin et al., 2019), which requires no labeled training data and can categorize each news item into our predefined narrative classes. We avoid using sentiment lexicons or self-trained word embeddings/classifiers because the former only match isolated words without considering context or negation, while the latter tend to overfit under conditions of limited labeled data and few available days per stock. Pre-trained Transformers, in contrast, naturally incorporate external knowledge at no additional cost. Based on this pipeline, our contributions can be summarized as follows.

First, in terms of methodology, this paper proposes Narrative-Gated Local Projections (NG-LP): each news article is first decomposed into multiple narrative dimensions, and only narratives that pass through both a “credibility” gate and a “recent relevance” gate are allowed to participate in voting. The final prediction for the next-day direction of individual stocks is determined by

daily majority voting. This design differs from the mainstream approach in existing literature — typically based on “overall sentiment plus portfolio-level analysis” — by instead modeling at the individual stock level with interpretable narrative structures. Second, empirically, we find that, under these constraints, NG-LP consistently outperforms the baseline out-of-sample in 2025 (outperforming by approximately 11 percentage points when combining the three core stocks), while more flexible models such as decision trees, SVM, LSTM, and logistic regression all suffer from overfitting and revert to baseline levels — what truly delivers gains is the combination of a simple linear model with narrative gating, rather than model capacity. Third, this study uses controlled experiments to delineate the signal’s boundaries: weak AI stock GOOGL systematically fails to outperform the baseline, indicating that this predictive power is specific to AI narratives rather than a general market effect; meanwhile, prediction accuracy follows a U-shaped distribution relative to realized volatility, suggesting the signal is effective only when markets exhibit clear directional trends. Fourth, the paper tests excess accuracy at both individual stock and aggregated levels using bootstrap methods: each core stock beats its own baseline with a bootstrap probability of approximately 90%, constituting marginally significant supporting evidence, while after pooling into the three core stocks this probability rises to 99%, reaching full significance; the consistency across both levels indicates that the result does not stem from chance in any individual stock. Fifth, regarding theoretical implications, the above evidence is in tension with the semi-strong form of the efficient market hypothesis — news narratives appear to retain lagged directional information not yet instantaneously priced in; however, the conclusions of this paper are strictly confined to directional predictability and do not claim that tradable excess returns can be obtained.

The remainder of this paper is organized as follows. Section 2 introduces data and feature construction, including sources of price and news data, sample period and selection criteria, as well as a two-step pre-trained language model pipeline for transforming news text into sentiment and narrative features. Section 3 characterizes the data environment faced by modeling, demonstrating that time series alone are insufficient for directional prediction through ACF/PACF analysis, volatility assessment, VAR spillover tests, and GSADF bubble detection. Section 4 derives the gating design of NG-LP from a series of failed baseline models. Section 5 presents empirical results, while Section 6 discusses implications, limitations, and future directions regarding the semi-strong form efficient market hypothesis.

2 Data

2.1 Data Sources and Grouping

This study uses two types of data: stock prices from Yahoo Finance, including daily closing prices and returns for five U.S. technology stocks (NVDA, MSFT, META, AMZN, GOOGL) from 2021 to 2025; and news data from Bloomberg, where we scraped web pages to extract the date, title, and summary of each article, totaling approximately 13,106 entries. Due to parsing limitations, the accuracy of matching dates with text is about 94%–95%. Since such mismatches are random rather than systematic, and NG-LP aggregates multiple news items via daily majority voting combined with dual gating mechanisms, individual mismatches are

averaged out during aggregation, resulting in limited impact on the conclusions.

Before examining the results, we pre-grouped the five stocks into two categories based on their prior association with AI. The core three stocks (NVDA, MSFT, META) have a strong connection with AI. The other two, AMZN and GOOGL, have a broader business scope and a weaker connection with AI. Therefore, we use these two as the control group (see Section 5).

2.2 From News Text to Features

Converting news into numerical form sets the ceiling for the quality of the entire pipeline. We abandoned the sentiment lexicon approach: even when adapted specifically for finance (Loughran & McDonald, 2011 demonstrated that general-purpose dictionaries misclassify financial terms and proposed a specialized financial lexicon), word-by-word matching fundamentally fails to handle context dependency, systematically losing meaning in sentences such as “demand has not slowed down,” which involve negation and contextual nuance. Instead, we adopted a two-step pretraining pipeline, allowing each model to fulfill its distinct role.

Sentiment polarity — FinBERT (Araci, 2019). FinBERT is a BERT model fine-tuned on financial corpora, capable of accurately understanding financial terms such as “beat,” “miss,” and “downgrade” along with their negations. For each news article, the model outputs probabilities for positive, negative, and neutral sentiments. We define the sentiment score as $\text{score} = P(\text{pos}) - P(\text{neg})$, and confidence as $\text{conf} = \max(P(\text{pos}), P(\text{neg}))$, which is used later for gating.

Narrative classification — BART zero-shot (Lewis et al., 2020; Yin et al., 2019). We lack labeled news data, and the number of available days per stock is limited, making self-trained classifiers prone to overfitting. Therefore, we adopt BART’s zero-shot classification: each news article is independently scored against five candidate narrative labels (multi-label, each score ranging from 0 to 1), and any label exceeding a threshold of 0.6 is assigned. The threshold of 0.6 is based on practical experience — only approximate categorization is needed, not fine-tuning.

The five narrative categories represent a design choice involving subjective judgment. The five categories — chips, power, algorithms, regulation, and financial reports — were manually defined by us based on our understanding of the AI industry chain. It must be emphasized that this classification itself embeds the researcher’s investment perspective (echoing Shiller’s (2017) narrative economics view, in which narratives themselves are economic forces driving prices). Because different category definitions produce different signals, this choice is an active modeling decision, not a neutral preprocessing step. Our narrative-by-narrative approach is also in line with recent work: Chen, Kelly & Xiu (2023) clustered news into topics using LLM embeddings and likewise found that predictive strength varies substantially across themes. This is both part of our methodology and its limitation — how to enable models to autonomously discover and select narrative categories remains a direction we explore in Section 6.

3 Preliminary Data Analysis

Before building a news-based predictor, it is essential to first examine the data from a purely statistical perspective: if stock price movements can be adequately captured by historical price trends or inter-stock correlations, incorporating news would be redundant. This section uses

three sets of tools to address this question, and their consistent conclusion is that predicting direction based solely on time series data is nearly impossible — this very insight forms the starting point for our shift toward news narratives.

3.1 Return Series, Autocorrelation, and Volatility

The daily return series of the equal-weighted portfolio is shown in Figure 1, which visually appears as noise without any obvious structure. An ACF and PACF analysis of the daily returns reveals that lag coefficients at all orders generally fall within the 95% confidence bands, indicating no exploitable linear autocorrelation in returns. However, volatility behaves differently: the ACF/PACF of the 5-day and 21-day rolling volatilities (Figure 2) exhibit significant and persistent autocorrelation — typical volatility clustering. In other words, only the magnitude of volatility is predictable, not the direction of returns — a disadvantageous starting point for tasks aimed at predicting price increases or decreases.

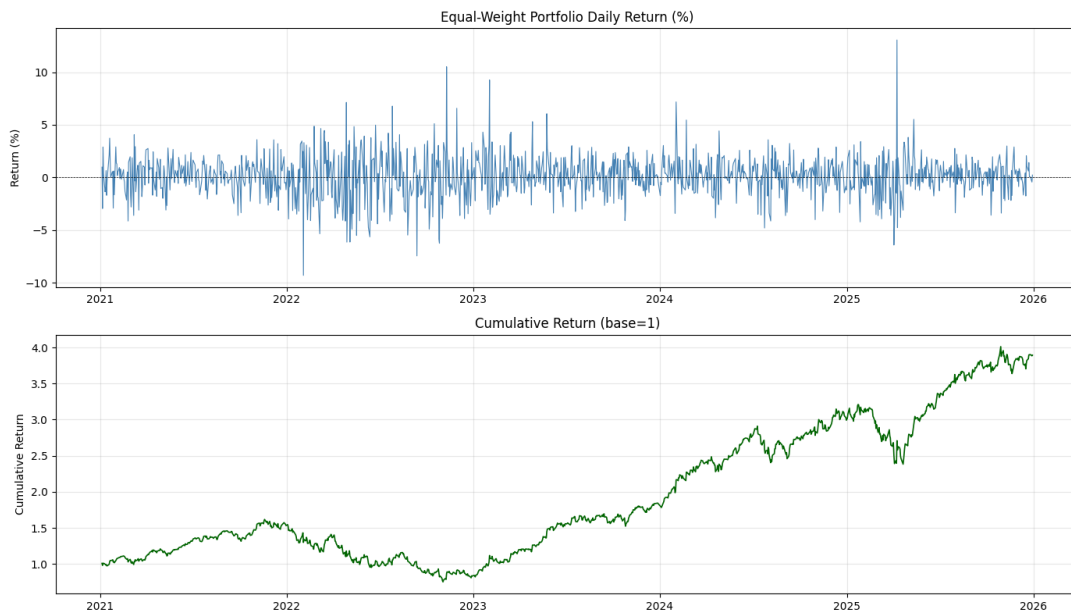


Figure 1: Daily returns (top) and cumulative returns (bottom) of the equal-weighted portfolio, 2021–2025.

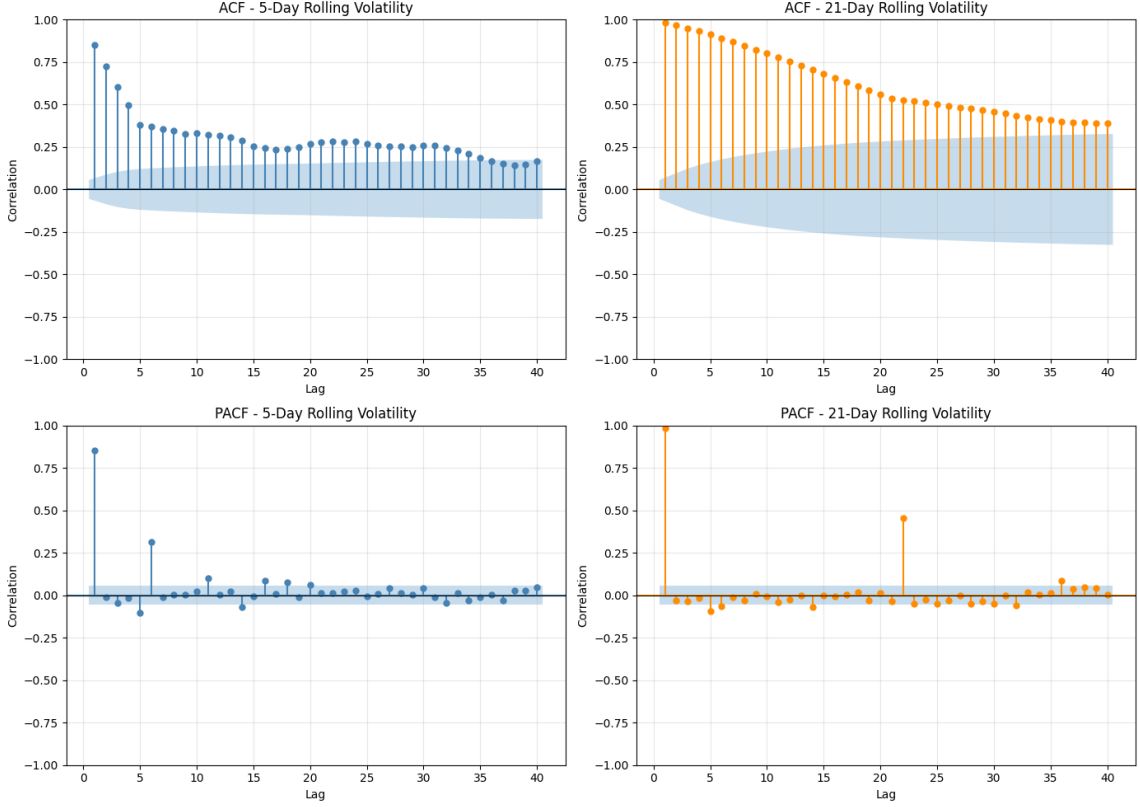


Figure 2: ACF and PACF of 5-day and 21-day rolling volatility: significant and persistent autocorrelation (volatility clustering).

3.2 Stock Interconnectedness: VAR Spillover

To examine whether the direction of individual stock returns is led by historical information from other stocks, we fit a vector autoregression (Sims, 1980) to stock returns and conduct Granger causality tests, with the core specification being

$$r_t = c + \sum_{k=1}^p A_k r_{t-k} + \varepsilon_t, \quad (1)$$

where r_t is the return vector and A_k is the coefficient matrix. We estimate separately on five stocks (Figure 3) and the three AI core stocks (Figure 4). The impulse responses for both groups show that cross-stock causality is generally insignificant, and among the three core stocks, there is almost no significant lead-lag relationship in 2025 — each stock is primarily driven by its own information, and the cross-stock structure provides no directional cues.

IRF — 5-stock close-close, full period 2021-2025 (n=1253, p=1)

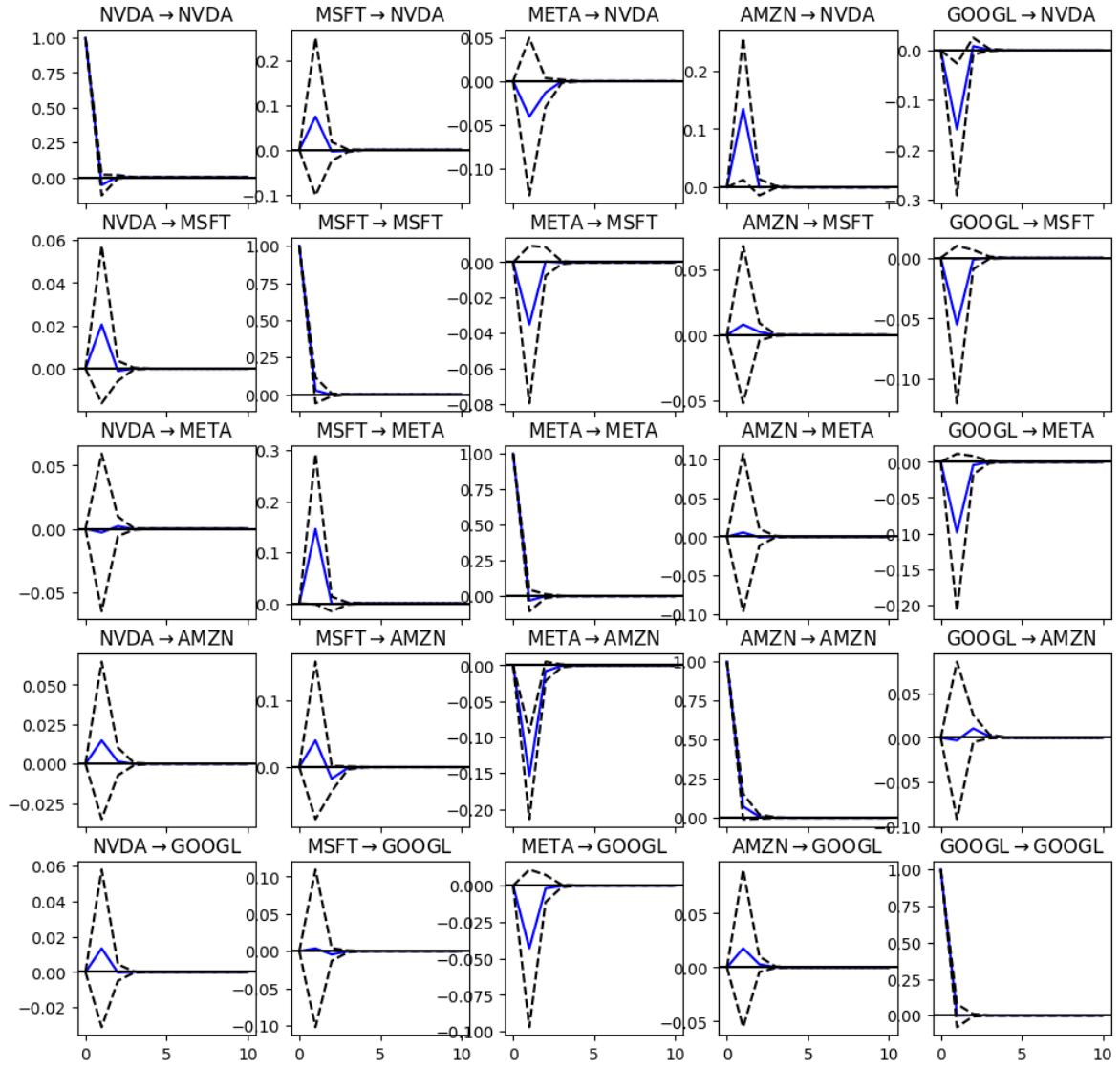


Figure 3: Impulse response (IRF) of the five-stock VAR, 2021–2025: cross-stock responses generally fall within the confidence bands.

IRF — 3-stock (no GOOGL/AMZN) close-close, 2025 (n=249, p=1)

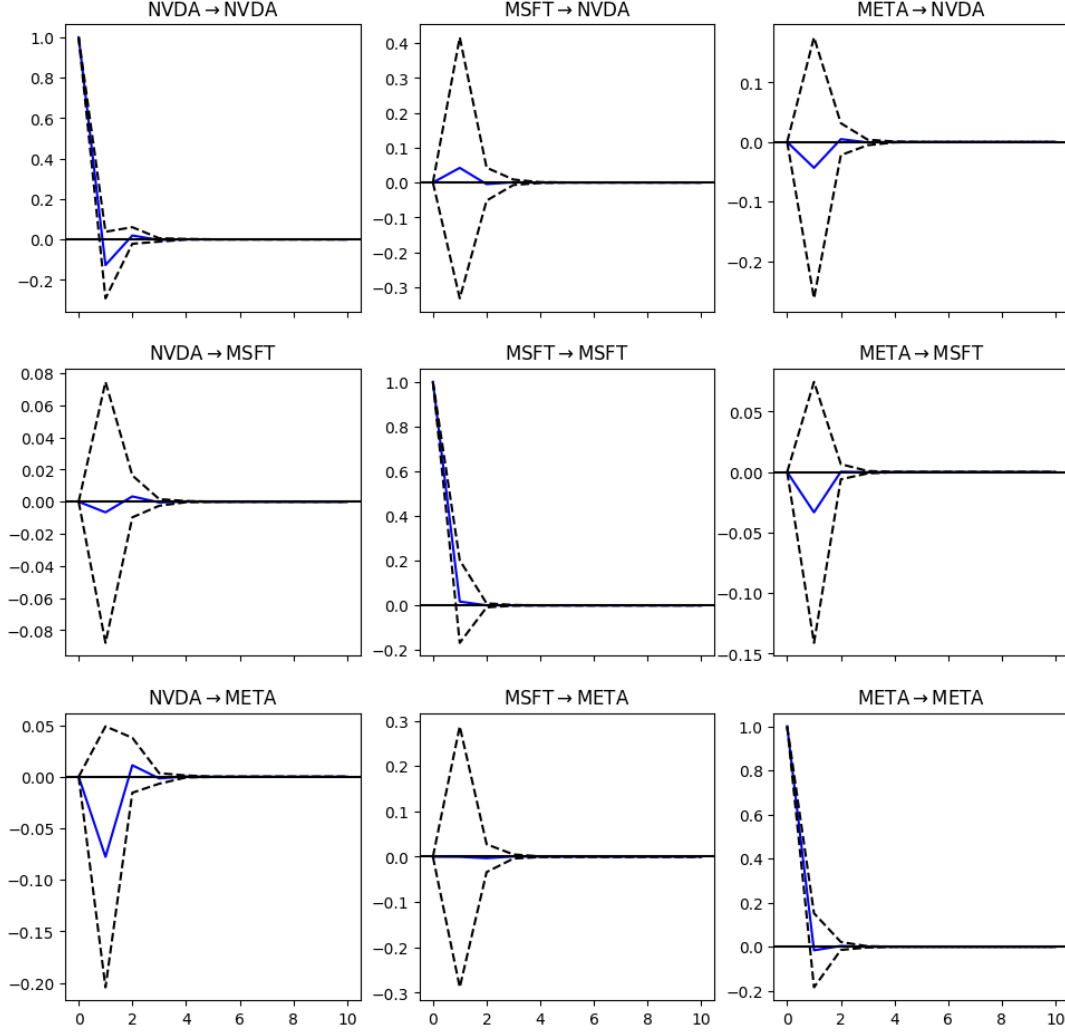


Figure 4: Impulse response (IRF) of VAR for the three AI core stocks (NVDA/MSFT/META): cross-stock causality remains insignificant.

3.3 Explosive Dynamics: GSADF Bubble Test

We use the GSADF test (Phillips, Shi & Yu, 2015) to characterize explosive behavior in prices, which takes the supremum of right-tailed ADF statistics over rolling subsamples:

$$\text{GSADF} = \sup_{r_1, r_2} \text{ADF}_{r_1}^{r_2}. \quad (2)$$

The results (Figure 5) indicate that the bubble is stock-specific rather than market-wide: NVDA experienced a significant surge (GSADF = 2.20, with the bubble period concentrated in 2023–2024), MSFT showed no evidence of a bubble (GSADF = 0.64), META and GOOGL exhibited moderate signals, while for AMZN the null hypothesis was rejected but without substantial bubble episodes. These explosive episodes are concentrated in time within individual stocks yet scattered across different names, and thus do not constitute a common signal exploitable across stocks.

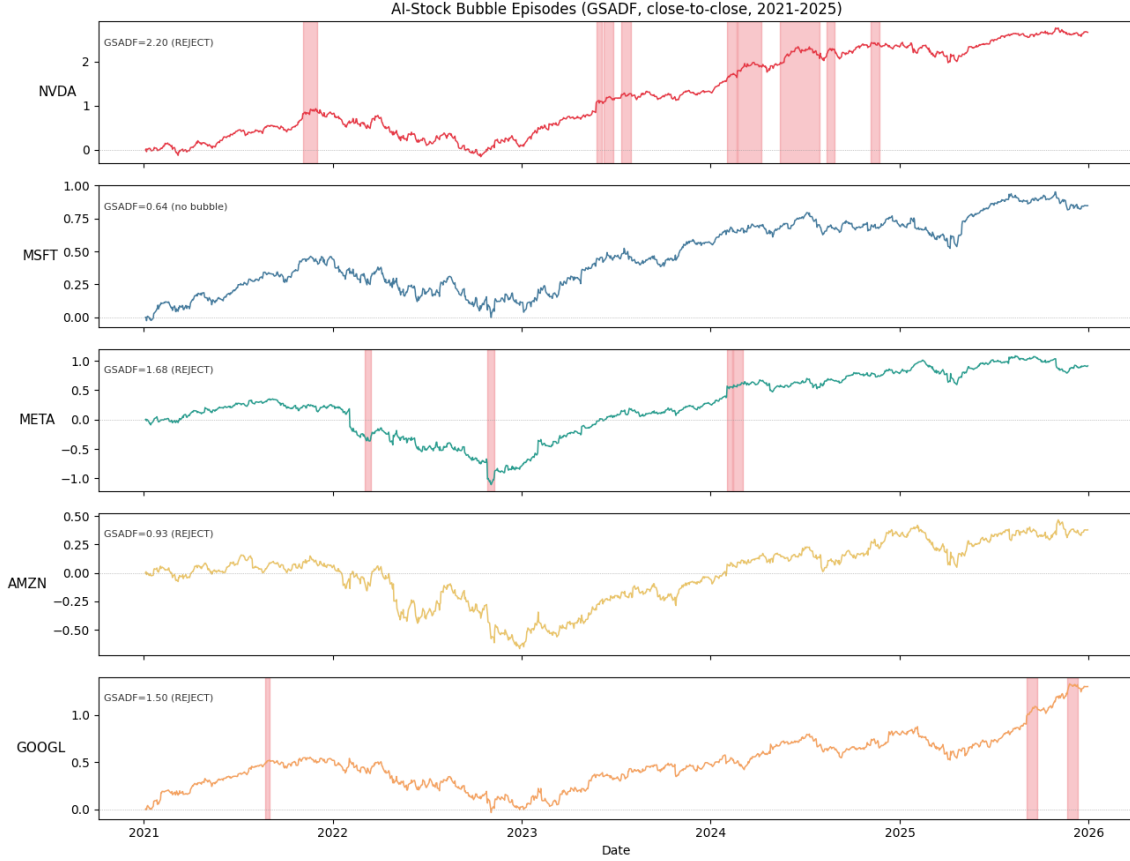


Figure 5: GSADF bubble detection: shaded areas indicate explosive periods, with bubbles being stock-specific and dominated by NVDA.

3.4 Summary

Three sets of analyses point to the same conclusion: returns exhibit no linear autocorrelation, cross-stock correlations are insignificant, and bubble stocks are idiosyncratic and scattered. The time-series structure of prices itself contains almost no information useful for predicting direction. If any predictability in direction still exists, it must stem from factors outside price — leading us to news narratives.

4 From Failed Baselines to NG-LP

This chapter traces the reasoning chain that led to NG-LP through a series of structured failures: each model fails in a specific way, and every failure points toward the next improvement. We sequentially evaluate decision trees, SVMs, logistic regression, LSTMs, and local projection on this same dataset, ultimately converging on the gated design. Every model uses the same protocol: we train on 2021–2024, test out-of-sample on 2025, and compare each against a majority-class baseline that always predicts whichever direction was more common during the training period.

4.1 Bagging Decision Trees

We begin with a flexible nonlinear learner. For each stock we build a 7-dimensional feature vector from the five gated narrative sentiment means together with the mean FinBERT sentiment score and its average confidence, and use it to predict the next day’s direction. The learner is a bagged ensemble of decision trees fitted on bootstrap samples. What comes back is textbook overfitting (Table 1): all five stocks train above 0.6 accuracy, yet every one of them falls below its baseline out-of-sample, the average gap between training and out-of-sample accuracy is around 13 points, and no stock’s confidence interval reaches its baseline. The reason is simple. With only a few hundred trading days per stock, a model this flexible ends up memorising noise instead of learning anything that generalises. That is the first lesson of the chapter: when samples are small and signals weak, flexibility hurts more than it helps.

Table 1: Out-of-sample performance of bagged decision trees (training 2021–2024, testing 2025, threshold 0.6).

Stock	Train acc.	Test acc. (2025)	Baseline	Test – Baseline	Overfit gap
NVDA	0.667	0.523	0.539	−0.016	+0.143
MSFT	0.674	0.487	0.563	−0.076	+0.187
META	0.589	0.508	0.516	−0.008	+0.081
AMZN	0.606	0.473	0.511	−0.038	+0.133
GOOGL	0.645	0.524	0.543	−0.019	+0.121

4.2 Support Vector Machine (SVM)

Decision trees fail because they follow the data too eagerly, treating random wiggles in the training set as if they were real patterns. An SVM works differently. It looks for a clean separating boundary between up and down days and keeps a wide margin around it, so that a few stray points cannot pull that boundary out of shape. As a result, it naturally avoids treating noise as a pattern. We tested two approaches using SVM. The first approach aggregates daily news into a single numerical vector for each day, then uses SVM to directly predict whether the market will rise or fall on that day. The second approach assigns a score to each individual news item, sums these scores by day, and determines the market direction based on which side receives more votes. Both models use identical settings, differing only in whether voting occurs at the “news level” or judgment at the “daily level.”

After switching to SVM, overfitting was significantly alleviated — the gap between training and testing accuracy narrowed considerably, with approach two showing almost no difference. However, eliminating overfitting did not improve predictive performance: regardless of the aggregation method or gating threshold used, out-of-sample accuracy fluctuated only around its respective baseline. In all 30 configurations (5 stocks $\times$ 3 thresholds), none of the lower bounds of the confidence intervals exceeded the baseline, and even slightly higher results fell within the bootstrap error range.

This result indicates that the issue goes beyond model flexibility: even when overfitting is controlled, using a single model to make overall judgments across five narrative types still fails to isolate the signal. The next section will instead model each narrative type separately to test

whether the signal has been obscured by this holistic approach.

Table 2: Out-of-sample accuracy of the two SVM approaches (tested on 2025, threshold 0.6).

Stock	Baseline	Approach 1 (daily-level)	Approach 2 (news-level voting)
NVDA	0.539	0.547	0.477
MSFT	0.563	0.597	0.437
META	0.516	0.461	0.516
AMZN	0.511	0.489	0.489
GOOGL	0.543	0.543	0.533

4.3 Logistic Regression

Both trees and SVMs are flexible models, and their failure might be attributed to overcapacity. To rule out this explanation, we revert to the simplest linear setting. First, apply a double 0.6 gating: retain only news items i with confidence $\text{conf}_i = \max(P_i(\text{pos}), P_i(\text{neg})) \geq 0.6$; for narrative class c , take the signed sentiment value $s_{i,c}$ only if BART assigns the news item to that class ($\text{bart}_{i,c} \geq 0.6$), otherwise set it to 0. Aggregate these values by stock and by day to obtain the five-dimensional narrative feature vector $x_t = (x_{t,1}, \dots, x_{t,5})$ for each day. Based on this, we fit a logistic regression model separately for each stock:

$$\Pr(y_t = 1 \mid x_t) = \sigma\left(\beta_0 + \sum_{c=1}^5 \beta_c x_{t,c}\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (3)$$

where $y_t = \mathbf{1}\{r_{t+1} > 0\}$ indicates the next day’s price movement. The prediction for the current day is made based on $\Pr \geq 0.5$, and the direction is determined by daily net voting. This model contains no nonlinear terms and has only six parameters per stock, leaving virtually no room for overfitting in theory.

The results confirmed this point, yet fell short once again. The coefficients β_c for individual stocks were mostly insignificant, indicating that there is no stable global linear mapping between the five narrative types and market direction. Out-of-sample performance closely followed the baseline: among the five stocks, only MSFT showed a significantly higher accuracy than its baseline (0.608 vs. 0.541), but its confidence interval lower bound did not exceed the baseline, so it cannot be considered robust evidence; the other four remained near or below the baseline. In other words, logistic regression confirmed in the most conservative way what SVM had already suggested — the issue lies not in overfitting, but in using a single, uniform linear relationship shared across all stocks to model the connection between news and market direction, an assumption inherently too coarse.

Table 3: Out-of-sample performance of logistic regression (Test 2025).

Stock	Days	Test acc. (2025)	Baseline	95% CI
NVDA	111	0.505	0.541	[0.414, 0.595]
MSFT	74	0.608	0.541	[0.500, 0.716]
META	79	0.506	0.519	[0.392, 0.620]
AMZN	64	0.547	0.516	[0.422, 0.672]
GOOGL	74	0.500	0.568	[0.392, 0.608]

However, logistic regression also leaves a useful clue: the coefficients of a few “stock-narrative” combinations are close to being significant, suggesting that the signal may be narrative-specific and stock-dependent, rather than captured by a single unified coefficient. This clue will be formally exploited in the local projection method at the end of this chapter.

4.4 Long Short-Term Memory Network (LSTM)

Previous models have only considered news features from the current day. However, a natural question arises: does the effect of narratives accumulate over time — could a particular topic continue to develop over several consecutive days, with its impact only becoming apparent across a sequence of days? If so, a sequential model with memory should be able to capture this structure. Recent studies have attempted to combine news sentiment with LSTM for stock price prediction (e.g., FinBERT-LSTM approaches, Gu et al., 2024). Motivated by this, we introduce LSTM: it maintains a cell state at each time step and uses gating mechanisms to determine how much historical information to retain and how much current input to write in, thereby modeling dependencies across days.

Specifically, let the input on day t be $x_t \in \mathbb{R}^{11}$ (1 dimension for FinBERT sentiment + 5 dimensions for five narrative scores + 5 dimensions for stock dummy variables), and h_{t-1} be the hidden state at the previous time step. The update of the LSTM unit is as follows:

$$\begin{aligned}
f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f), & i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i), \\
\tilde{c}_t &= \tanh(W_c[h_{t-1}, x_t] + b_c), & o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o), \\
c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, & h_t &= o_t \odot \tanh(c_t),
\end{aligned} \tag{4}$$

where σ is the sigmoid function, $\odot$ denotes element-wise multiplication, $W_\bullet$ and $b_\bullet$ are the weights and biases to be estimated, and f_t, i_t, o_t represent the forget gate, input gate, and output gate, respectively. The model takes a sequence of length $K = 10$ ($x_{t-9}, \dots, x_t$) as input, and uses the final hidden state h_t to generate a directional prediction $\hat{y}_t = \sigma(w^\top h_t + b)$ via a fully connected layer with a sigmoid activation (Figure 6). The hidden dimension is set to 16, with dropout (0.3) and weight decay applied; we use 2021–2023 for training, 2024 for validation with early stopping, and 2025 for out-of-sample testing.

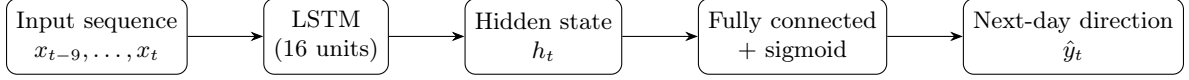


Figure 6: Schematic of the LSTM direction-prediction structure: inputting a 10-day feature sequence, extracting the final hidden state, and using a fully connected layer to predict the next day’s price movement.

Contrary to the expectation that “larger models are better” (Table 4), the training accuracy reached 0.631, while the validation accuracy in 2024 was only 0.532 and the out-of-sample accuracy in 2025 dropped to 0.522 — slightly below the baseline of 0.533. An 11-point gap between training and test performance is a clear sign of overfitting. Adding memory, more capacity, and regularisation (dropout, weight decay, early stopping) changed nothing: the LSTM still could not find any cross-day narrative signal. The lesson repeats: when data is this scarce, a bigger model just overfits harder.

Table 4: LSTM accuracy at each stage (combined training, by year).

	Train (2021–23)	Validation (2024)	Test (2025)	Baseline
Accuracy	0.631	0.532	0.522	0.533

4.5 Local Projections: One Narrative at a Time

The clue left by logistic regression is that signals may be narrative-specific and vary across stocks. Based on this insight, we completely change the modeling granularity — instead of using a single model to capture all narratives, we estimate separately for each “stock-narrative” combination. For narrative category c of a given stock, we perform a simple linear regression of next-day returns on the sentiment value of that narrative on the current day. This is precisely the form of local projection (Jordà, 2005):

$$r_{t+1} = \alpha_c + \beta_c s_{t,c} + \gamma_c r_t + \varepsilon_{t+1}, \quad (5)$$

where $s_{t,c}$ is the signed sentiment of narrative c on that day, and r_t is the daily return used as a control variable. The sign of coefficient β_c indicates the narrative’s prediction for the next day’s direction: if a narrative has positive sentiment on a given day and $\beta_c > 0$, it casts a vote for “up.” On each trading day, all active narratives cast one vote, and the majority determines the directional forecast for that day.

This is the first setting where the out-of-sample performance truly exceeds the baseline (Table 5). Without any gating, NVDA achieves an out-of-sample accuracy of 61.5%, 5.5 percentage points higher than the baseline, while MSFT reaches 58.3%, 4.8 percentage points above the baseline. The signal demonstrates its reference value for the first time — linear, narrative-driven, and requiring no complex models.

Table 5: Out-of-sample performance of ungated local projection (Test 2025): first occurrence of positive excess.

Stock	Accuracy	Baseline	Excess	Days	$P > \text{base}$
NVDA	61.5%	56.0%	+5.5%	96	88%
MSFT	58.3%	53.6%	+4.8%	60	73%

But it is not yet stable: on META and the remaining stocks the same setup comes out flat, sometimes negative. The problem is that a stock’s investment logic does not stay put. A narrative that drives prices in one stretch can go quiet, or even flip sign, in the next. If we estimate β_c once on 2021–2024 and then apply it blindly across all of 2025, we are bound to get it wrong on the days when the logic has already changed. What we need is a way to track this drift and to trust a narrative only while it is actually working. That is what NG-LP does.

4.6 NG-LP: Narrative-Gated Local Projection

The previous section pointed the way: the per-narrative local projection is the only setup that has produced a real, significant signal, so we keep it as the backbone. What it lacks is stability, and we think the reason is drift. A stock’s investment logic shifts over time, so a narrative can be informative in one period and dormant or reversed in the next; voting on every day alike lets stale signals dilute the ones that currently matter. NG-LP keeps the LP backbone but decides, day by day, which narratives are allowed to speak. In one sentence: on each trading day we re-estimate every narrative inside a sliding window, let only the narratives that clear a two-stage gate vote, and take that day’s majority as the next-day direction (Figure 7). It has three layers, each fixing one problem.

(1) Sliding window: handling logic drift. Each day, a narrative’s sentiment loading β is re-estimated using only its most recent K “signal days”, always rolling forward on past data. This keeps the model tuned to the current market logic instead of freezing one estimate over the whole sample. We use $K = 30$ as the main setting and $K = 50$ for robustness checks.

(2) Significance gate: the entry exam (Gate 1). A narrative may vote only if its β is statistically significant in the current window. To check this, we draw 400 bootstrap resamples from the window, rebuild the distribution of β , and take its 5th and 95th percentiles as a two-sided 90% confidence interval; the narrative passes only if that interval excludes zero. In effect, this gate lets a narrative speak only when the “news–return” link is really there, and screens out pure noise.

(3) Performance gate: the probation period (Gate 2). Being statistically significant does not guarantee getting the next day right, so we add a finite-state machine that tracks each narrative’s recent hits and misses: three misses in a row and it is benched; it only comes back after two hits in a row. A narrative that has stopped working is demoted until it earns its place again. The two gates play complementary roles: the significance gate checks whether the relationship holds in-sample, and the performance gate checks whether it still pays off out-of-sample.

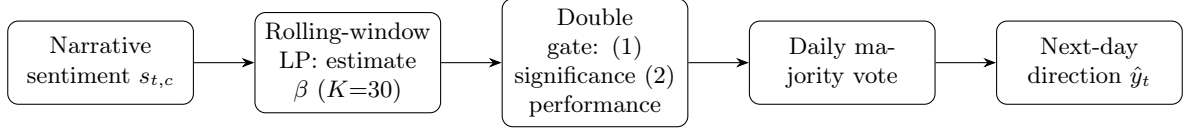


Figure 7: NG-LP process illustration: sliding-window local projection estimates each narrative loading, which is then filtered through dual gates of significance and performance, with the majority vote per day determining the direction for the next day.

Forward-looking estimates with strict no-look-ahead guarantee. For each narrative category, we advance in chronological order on its active signal days ($|s_{t,c}| > \text{SENT_MIN}$, where $\text{SENT_MIN} = 0.05$). On day i , we take a window $w = \{i - K, \dots, i - 1\}$, fit a local projection via least squares over this window to obtain the loading β , and predict “up $\Leftrightarrow \beta \cdot s_{i,c} > 0$ ”. The entire process is strictly free of look-ahead: estimation windows are left-closed and right-open; the predicted day i never enters its own estimation window; all target values within the window have already been realized before day i ; predictions use only sentiment known on that day; the baseline adopts a point-in-time approach, averaging returns strictly prior to the prediction date — thus the 2025 test set never sees its own outcomes; the state machine updates using only observed past hits or misses.

Vote aggregation and pooling across stocks. Inside a stock, every narrative that clears both gates casts one vote that day, the majority sets the stock’s single direction for the day, and ties are dropped. Each surviving day carries its own point-in-time baseline; grouping by year, we evaluate accuracy, excess return, and $P(\text{acc} > \text{base})$ with 5,000 bootstrap samples. Here $P(\text{acc} > \text{base})$ is the fraction of these resamples — drawn by resampling the stock-day observations with replacement — in which NG-LP’s accuracy exceeds the baseline; it is a resampling frequency that summarises how stable the excess is, not a frequentist p -value from a formally specified null. Pooling across stocks is deliberately simple: we stack each stock’s daily votes into one table without re-voting across stocks, so every stock keeps its own daily majority and each “stock-day” counts as one observation in a single, larger accuracy figure. The point is statistical power: a single stock has only a few dozen voting days a year, too few for precise significance, whereas stacking independent daily calls gives a much sharper test. We report two pools, all five stocks together and the set excluding GOOGL.

Where the hyperparameters come from. Every hyperparameter is fixed from prior reasoning or the 2024 validation period, never tuned on the 2025 test set: the 0.6 gate is the lowest confidence we are willing to call “credible”; $K = 30$ reflects the assumption that narrative logic turns over roughly monthly; the 3/2 rule in the FSM follows from a probability argument under a 50% baseline; and $\text{SENT_MIN} = 0.05$ drops the near-neutral days.

5 Empirical Results

5.1 Core Three Stocks’ Out-of-Sample Performance in 2025

We present the main results on the AI core trio (NVDA, MSFT, META), a group predetermined prior to observing the outcomes. Here we adopt an intentionally stricter baseline: the actual probability of positive return for each stock over the full year of 2025, $P(r_{t+1} > 0)$, rather than

a point-in-time baseline. Since 2025 was an up year, its realized return rate (approximately 53.5%) exceeds that of earlier periods with lower returns, making outperformance in 2025 a more conservative conclusion.

The results are shown in Table 6. After aggregating 107 “stock-day” observations, the out-of-sample accuracy of NG-LP reached 64.5%, compared to a baseline of 53.5%, representing an excess gain of 11.0 percentage points. The 95% bootstrap confidence interval was [55.1%, 72.9%], with $P(\text{acc} > \text{base}) = 99\%$ and $P(\text{acc} > 50\%) = 100\%$. To move beyond a resampling frequency and obtain a formally specified test, we additionally conduct a permutation test: the realized next-day directions are randomly reshuffled across the pooled voting days (preserving the marginal up-rate) and the accuracy is recomputed, yielding a null distribution centred at 51.9%. Across 20,000 permutations, NG-LP’s 64.5% accuracy is exceeded in only 0.6% of cases, giving a one-sided p -value of $p = 0.006$. Unlike $P(\text{acc} > \text{base})$, this is a genuine test against the null of no directional information, and the pooled result clears it comfortably. At the individual stock level, each core stock outperformed its own baseline with at least 90% bootstrap probability. This provides marginally significant supportive evidence at the individual level, while the aggregated result achieves full significance. The consistency across both levels indicates that the results are not due to random coincidences on specific trading days for particular stocks, but rather attributable to the overall sample admitted. It should be noted that predictions for the three core stocks on the same trading day are correlated, so the effective number of independent observations in the pooled sample is less than the nominal 107. Therefore, the 99% significance at the pooled level should be interpreted conservatively. To gauge how much this dependence actually inflates our confidence, we re-run the bootstrap by resampling whole trading days rather than individual stock-days (a cluster bootstrap that keeps each date’s observations together). The 107 stock-days in fact span 90 distinct trading dates, so the overlap is modest: the cluster-bootstrap 95% interval is [54.7%, 73.5%] with $P(\text{acc} > \text{base}) = 99\%$, essentially unchanged from the naive interval. The correlation across same-day core-stock predictions therefore does not materially weaken the pooled evidence. Taken together, the permutation test and the cluster bootstrap indicate that the pooled evidence is the stronger of the two, while the individual-stock results remain consistent but only marginally significant.

Table 6: Out-of-sample performance of NG-LP on the core three stocks in 2025.

Stock	Days	Accuracy	Baseline '25	Excess	$P > \text{base}$	$P > 50\%$
NVDA	49	63.3%	53.6%	+9.6	91%	97%
MSFT	27	66.7%	54.4%	+12.2	91%	96%
META	31	64.5%	52.4%	+12.1	90%	95%
Core three (pooled)	107	64.5%	53.5%	+11.0	99%	100%

5.2 Control Experiment: Is the Signal Unique to AI Narratives?

If the above predictive power truly stems from AI narratives rather than some broad market effect, it should fail on weak-AI stocks, and removing such stocks should enhance — rather than weaken — the overall result. Pooling tests support this conclusion (Table 7). With all five pooled, we get 60.0% out-of-sample accuracy in 2025 (+7.1% excess); dropping GOOGL raises

this to 64.5% (+11.9%). Removing a stock therefore helps, which means GOOGL dilutes the signal rather than adding to it. GOOGL on its own confirms this: 40.0% accuracy, 13.7 points below its own baseline — a clear systematic failure. Driven primarily by advertising, GOOGL does not have AI as the central narrative underpinning its price movements, meaning narrative gating has no leverage here — this very failure serves as evidence that the signal is specifically tied to AI narratives. (Another weak-AI stock, AMZN, had only three voting days passing the gate in 2025, too sparse a sample to draw any conclusions.)

Table 7: Pooling and control experiment (out-of-sample, 2025).

Pool	Days	Accuracy	Baseline	Excess	$P > \text{base}$
All five	135	60.0%	52.9%	+7.1	96%
Excluding GOOGL	110	64.5%	52.7%	+11.9	99%
GOOGL alone	25	40.0%	53.7%	−13.7	9%

5.3 Robustness of Window Length

The main results use a sliding window of $K = 30$. To verify that the conclusions are not dependent on this specific choice, we re-estimated the results by expanding the window to $K = 50$. The core narrative signals still outperformed the baseline: NVDA’s chip narrative achieved an accuracy of 66.7% in 2025, with an excess return of +13.0%; MSFT’s power narrative delivered +4.9%, and META’s power narrative +5.5%. The key narratives for these core stocks remain effective under a longer window, indicating that the main findings are not artifacts of the $K = 30$ window length.

5.4 Volatility Layering: U-Shaped

We split the 107 pooled trading days of the core three into three bands by the realised volatility of the following day, $|r_{t+1}|$, and the pattern is clearly U-shaped (Table 8). Both the low and high bands reach 72.2% accuracy ($P > \text{base} = 99\%$), while the middle band manages only 48.6% ($P > \text{base} = 31\%$), barely better than a coin flip. The model therefore does best when the market has a clear direction, whether that means quiet, small moves or violent, sharp ones; it is the murky middle, where direction is ambiguous, that news narratives cannot read. In short, NG-LP is a conditional predictor: it works when the market is trending clearly, and not much otherwise.

Table 8: Accuracy by next-day volatility tier (core three stocks combined, 2025).

Volatility band	Days	Accuracy	95% CI	$P > \text{base}$	Mean $ ret $
Low	36	72.2%	[58.3%, 86.1%]	99%	0.26%
Middle	35	48.6%	[31.4%, 65.7%]	31%	1.05%
High	36	72.2%	[55.6%, 86.1%]	99%	3.68%

5.5 Notes on 2026 Data

In the interest of transparency, we must acknowledge the temporal limitations of our findings. Extending the NG-LP model to data from 2026 onward faces two major obstacles. First, the

sample size available for 2026 is too small to support any meaningful significance estimates. Second, even with this limited data, the model’s accuracy does not exceed that of the baseline. Therefore, we cannot test the validity of this mechanism using 2026 data — neither confirming its continued relevance nor refuting it. Our empirical claims are strictly confined to the 2021–2025 sample period; whether they extend beyond 2025 remains an open question.

6 Discussion, Limitations, and Future Perspectives

6.1 Conclusion

On a sample of five U.S. AI technology stocks from 2021 to 2025, NG-LP consistently outperformed the baseline out-of-sample in 2025: the combined accuracy for the three core AI stocks reached 64.5%, compared to the baseline’s 53.5%, yielding an excess of +11.0 percentage points, with $P(\text{acc} > \text{base}) = 99\%$. Moreover, each individual core stock outperformed its own baseline. This suggests that news narrative structures carry information about the next day’s market direction. Regarding the Efficient Market Hypothesis, we adopt a tiered interpretation: at the descriptive level, the model decisively surpasses a deliberately stringent out-of-sample baseline; at the inferential level, this outperformance creates tension with full semi-strong form market efficiency — narrative structures appear to contain lagged directional signals that have not yet been fully priced and may be exploitable. However, we do not make the stronger claim of “attainable and tradable excess returns,” which remains for future research.

A recurring insight throughout the text is: why the simplest model ends up winning. On such limited data, decision trees, SVMs, and LSTMs all overfit and revert to baseline performance, while only simple local projections remain stable. This suggests that the source of predictive power is the gating structure itself, not model capacity or nonlinearity — only accepting narratives when they are both statistically significant and recently effective. When data is scarce and noisy, tightening structural constraints helps more than scaling up the model. This phenomenon that “simple models are not necessarily inferior to large models” is not isolated: Shobayo et al. (2024) also found in stock index sentiment prediction that logistic regression outperformed FinBERT and GPT-4.

6.2 Limitations

The conclusions of this paper are subject to several limitations. First is the uncertainty in time extrapolation: as discussed in the previous section, data for 2026 is both limited and underperforms the baseline, making it difficult to confirm whether the narrative-gating logic holds beyond 2025 — it may be a product of specific market conditions in that year. Second is the subjectivity of narrative categories: the five categories were manually defined based on our understanding of the AI industry chain, incorporating researchers’ prior judgments; different definitions would yield different signals. Third is sample size: even after pooling, the core three-stock assessment for 2025 rests on only about a hundred “stock-day” observations. Additionally, by using close-to-close returns, the analysis includes news already priced in during intraday trading into the next day, introducing a conservative bias toward actual signals. The model

also has a low R^2 , in line with earlier work (e.g., Tetlock, 2007): the signal is good for calling direction, not for predicting the size of returns.

6.3 Outlook

The most fundamental limitation among the above is the artificial definition of narrative categories. Although the gating logic in this paper can determine whether a particular narrative category is effective at a given time, it cannot answer which categories are inherently appropriate. A natural next step is to let the model find and choose narrative categories from the data itself, instead of our fixed, hand-picked set. Built on data-driven narratives, the gating logic would have a better chance of holding up across different periods and stocks, which would test whether the mechanism generalises beyond the 2025 sample.

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